

and sends the results back through the connection just established. This completes the loop.

Within this loop there are small important components in the cloud computing architecture. For example a central server does the bulk of administration work and takes care of traffic monitoring and resource allocation. Middleware is software that allows machines in the back end network to communicate with each other. As you must have realised, such a large network of devices must have an optimisation framework in place, else there will be massive idle time, power wastage, etc. There is a hi-fi term called server virtualisation, wherein a physical server is divided into a number of “virtual” servers, each running at maximum capacity. This technique reduces the needed for more physical servers in the cloud computing framework. Cloud computing as a framework is no rocket science. The genius of cloud computing does not lie in its framework but its scale, and the scale is increasing with time as research progresses.

Cloud computing you use

If you have ever attended a tech / geek conference on any computer technology it near impossible to miss out on cloud computing being brought up. Slowly and steadily cloud computing has made its way to mainstream IT and computer science. A recent research study states that almost half of the US's business enterprises use cloud computing to manage its operations. This is quite a staggering figure and it reinforces the belief that cloud computing deserves to move on from simply being a buzzword and take its rightful place in the limelight.

Although you may not be aware of cloud computing, but you've surely heard of Facebook! The Social networking giant makes use of cloud computing to store all of that user data that you and your friends faithfully upload every day. Instagram, otherwise also known as the “I love cats and food” app, uses cloud computing facilities to store millions of photos and process and apply filters as well. Cloud Computing is the tech secret behind how all these apps are able to store data and run software without downtime.

Essentially Cloud Computing has stamped its authority in all industries that require massive computing, storage or data handling. The world of High Performance Computing has greatly benefited due to the cloud computing rise. Scientists can now access supercomputing-like power at hourly rental prices. This gives scientists working in different parts of the country the freedom to run complex scientific simulations, calculations and visualisa-